WEEK – 8

2# Consider the following database for a banking enterprise.
BRANCH(branch_name:string , branch_city:string , assets:real)
ACCOUNT(accno:int , branch_name:string , balance:real)
DEPOSITOR(customer_no:string , accno:int)
CUSTOMER(customer_name:string customer_no:string, customer_street:string , customer_city: string)
LOAN(loan_number:int , branch_name:string , amount:real)
BORROWER(customer_no:string , loan_number:int)
a) Create ER Diagram and Schema diagram for the above system.
b) Create the tables by properly specifying the primary keys and foreign keys.
c) Enter at least five tuples for each relation.
d) Find all the customers who have at least two accounts at the Aligarh's branch.
e) Find all the customers who have balance more than 100000 in their accounts.
f) Demonstrate how you delete all account tuples at every branch located in a specific city

a)

* attached at end

b)

```sql
create table BRANCH(
    branch_name VARCHAR2(30) PRIMARY KEY,
    branch_city VARCHAR2(30),
    assets REAL
);

create table Account(
    acc_no INT PRIMARY KEY,
    branch_name VARCHAR2(30),
    balance REAL,
    FOREIGN KEY (branch_name) REFERENCES BRANCH(branch_name)
);
```

```sql
create table Customer(
    customer_no VARCHAR2(20) PRIMARY KEY,
    customer_name VARCHAR2(50),
    customer_street VARCHAR2(100),
    customer_city VARCHAR2(50)
);

create table Loan (
    loan_number INT PRIMARY KEY,
    branch_name VARCHAR2(50),
    amount REAL,
    FOREIGN KEY (branch_name) REFERENCES BRANCH(branch_name)
);
```

```sql
create table Depositor(
    customer_no VARCHAR2(20),
    acc_no INT,
    PRIMARY KEY (customer_no, acc_no),
    FOREIGN KEY (customer_no) REFERENCES Customer(customer_no),
    FOREIGN KEY (acc_no) REFERENCES Account(acc_no)
);

create table Borrower(
    customer_no VARCHAR2(20),
    loan_number INT,
    PRIMARY KEY (customer_no, loan_number),
    FOREIGN KEY (customer_no) REFERENCES Customer(customer_no),
    FOREIGN KEY (loan_number) REFERENCES Loan(loan_number)
);
```

c)

```sql
insert into BRANCH values
('Aligarh', 'Aligarh', 5000000),
('Delhi', 'Delhi', 8000000),
('Mumbai', 'Mumbai', 9000000),
('Chennai', 'Chennai', 6000000),
('Kolkata', 'Kolkata', 7000000);

insert into Account values
(101, 'Aligarh', 120000),
(102, 'Aligarh', 150000),
(103, 'Delhi', 90000),
(104, 'Mumbai', 200000),
(105, 'Chennai', 110000);
```

```sql
insert into Customer values
('C1', 'Amit', 'Street1', 'Aligarh'),
('C2', 'Rahul', 'Street2', 'Delhi'),
('C3', 'Sneha', 'Street3', 'Mumbai'),
('C4', 'Priya', 'Street4', 'Chennai'),
('C5', 'Karan', 'Street5', 'Kolkata');

insert into Loan values
(201, 'Aligarh', 50000),
(202, 'Delhi', 70000),
(203, 'Mumbai', 80000),
(204, 'Chennai', 60000),
(205, 'Kolkata', 90000);
```

```sql
insert into Depositor values
('C1', 101),
('C1', 102),
('C2', 103),
('C3', 104),
('C4', 105);

insert into Borrower values
('C1', 201),
('C2', 202),
('C3', 203),
('C4', 204),
('C5', 205);
```

Results | Explain | Describe | Saved SQL | History

5 row(s) inserted.

0.07 seconds

d)

```sql
SELECT D.customer_no
FROM Depositor D
JOIN Account A ON D.acc_no = A.acc_no
WHERE A.branch_name = 'Aligarh'
GROUP BY D.customer_no
HAVING COUNT(D.acc_no) >= 2;
```

e)

```sql
SELECT DISTINCT C.customer_name
FROM Customer C
JOIN Depositor D ON C.customer_no = D.customer_no
JOIN Account A ON D.acc_no = A.acc_no
WHERE A.balance > 100000;
```

f)

```sql
DELETE FROM DEPOSITOR
WHERE accno IN (
    SELECT accno FROM ACCOUNT
    WHERE branch_name IN (
        SELECT branch_name FROM BRANCH
        WHERE branch_city = 'Delhi'
    )
);

DELETE FROM ACCOUNT
WHERE branch_name IN (
    SELECT branch_name FROM BRANCH
    WHERE branch_city = 'Delhi'
);
```